IP1

A 4*4, 4bits/pixel original image is given by

6	13	12	13
12	6	7	12
13	7	7	12
14	11	11	14

(a) Apply full-scale contrast stretching to the image. Show your work and sketch the resulting image.

(b) Apply histogram equalization to the image. Show your work and sketch the resulting histogram.

If1 (a)

old min = 6

old max = 14

new min = 0

new max = $2^4 - 1 = 15$

$$\text{New value} = \frac{\text{Value} - \text{oldmin}}{\text{oldmax} - \text{oldmin}} \times (\text{new max} - \text{new min}) + \text{new min}$$

$$\frac{6-6}{14-6} \times (15-0) + 0 = 0$$

$$\frac{13-6}{14-6} \times (15-0) + 0 = 13$$

$$\frac{12-6}{14-6} \times (15-0) + 0 = 11$$

$$\frac{7-6}{14-6} (15-0) + 0 = 2$$

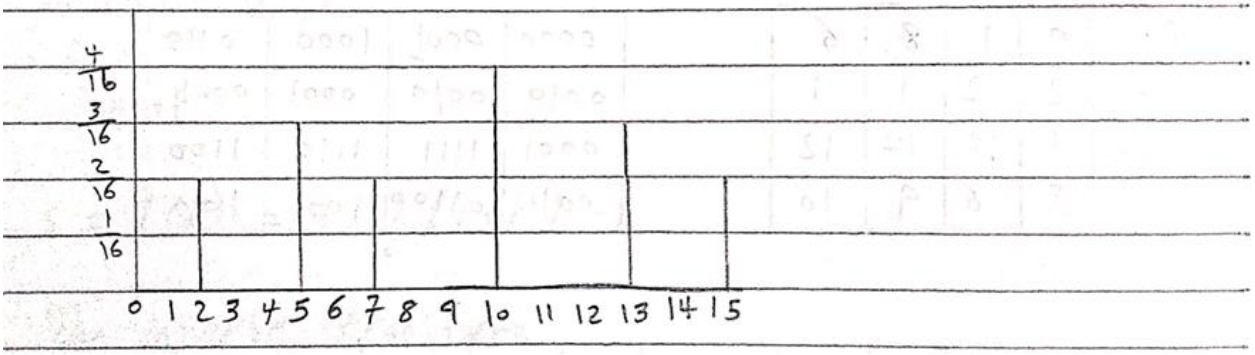
$$\frac{11-6}{14-6} (15-0) + 0 = 9$$

$$\frac{14-6}{14-6} (15-0) + 0 = 15$$

0	13	11	13
11	0	2	11
13	2	2	11
15	9	9	15

IP1 (b)

r_k	n_k	$P(r_k) = n_k/16$	cdf	$s_k = 15 \sum_{j=0}^{k} P(r_k)$	$P(s_k)$
6	2	2/16	2/16	2	2/16
7	3	3/16	5/16	5	3/16
11	2	2/16	7/16	7	2/16
12	4	4/16	11/16	10	4/16
13	3	3/16	14/16	13	3/16
14	2	2/16	1	15	2/16



IP2

255	13	127	7
55	40	150	255
160	20	200	14
170	90	255	9

255 = 11111111
13 = 00001101
127 = 01111111
7 = 00000111
55 = 00110111
40 = 00101000
150 = 10010110

160 = 10100000
20 = 00010100
200 = 11001000
14 = 00001110
170 = 10101010
90 = 01011010
9 = 00001001

0

1	1	1	1
1	0	0	1
0	0	0	0
0	0	1	1

1

1	0	1	1
1	0	1	1
0	0	0	1
1	1	1	0

2

1	1	1	1
1	0	1	1
0	1	0	1
0	0	1	0

3

1	1	1	0
0	1	0	1
0	0	1	1
1	1	1	1

4

1	0	1	0
1	0	1	1
0	1	0	0
0	1	1	0

6

1	0	1	0
1	1	0	1
1	0	0	0
1	0	1	0

6

1	0	1	0
0	0	0	1
0	0	1	0
0	1	1	0

7

1	0	0	0
0	0	1	1
1	0	1	0
1	0	1	0

IP4

For the image shown, find a transformation function (i.e. a look-up-table) that will change its histogram to match the one shown in Table 1. Draw the transformed image. Also determine the histogram of the transformed image. Assume that the processed images can only take integer values between 0 and 7 (including 0 and 7).

4	3	6	1	0	5
3	2	5	7	5	3
2	1	4	6	7	4
1	6	3	5	6	5
0	5	2	4	7	7

Gray level f	0	1	2	3	4	5	6	7
Histogram h(f)	8	6	4	2	2	3	3	2

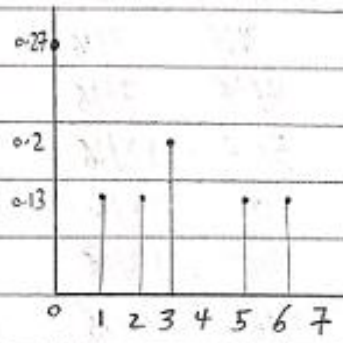
IP4

r	n	P(r)	cdf	s
0	2	$\frac{1}{15}$	$\frac{1}{15}$	0
1	3	0.1	$\frac{1}{8}$	1
2	3	0.1	$\frac{4}{15}$	2
3	4	$\frac{2}{15}$	0.4	3
4	4	$\frac{2}{15}$	$\frac{8}{15}$	4
5	6	0.2	$\frac{11}{15}$	5
6	4	$\frac{2}{15}$	$\frac{13}{15}$	6
7	4	$\frac{2}{15}$	1	7

z	n	P(z)	cdf	G(z)	s	G(z)	z
0	8	$\frac{4}{15}$	$\frac{4}{15}$	2	(0)	2	0
1	6	0.2	$\frac{7}{15}$	3	1	2	0
2	4	$\frac{2}{15}$	$\frac{9}{15}$	4	2	2	0
3	2	$\frac{1}{15}$	$\frac{10}{15}$	5	3	3	1
4	2	$\frac{1}{15}$	$\frac{11}{15}$	5	4	4	2
5	3	0.1	$\frac{5}{8}$	6	5	5	3
6	3	0.1	$\frac{14}{15}$	7	6	6	5
7	2	$\frac{1}{15}$	1	7	7	7	6

s	z
0	0
1	0
2	0
3	1
4	2
5	3
6	5
7	6

Z	P(Z)
0	0·27
1	0·13
2	0·13
3	0·2
4	0
5	0·13
6	0·13
7	0



2	1	5	0	0	3
1	0	3	6	3	1
0	0	2	5	6	2
0	5	1	3	5	3
0	3	0	2	6	6

Assignment

Assignment 1

- Assume that an image of 32x32 with 8 bit gray levels distributed as follow

Gray level	3	5	6	8	10	12	13	15
No. of pixels	124	50	100	245	150	275	30	50

- A)Draw the histogram, which type of images does we have?
- B)Draw the transformation function T(r),and then draw the equalized histogram

Assignment 2

- Q)Apply histogram equalization to the image. Show your work and sketch the resulting image.

6	13	12	13
12	6	7	12
13	7	7	12
14	11	11	14